📝 Unit 3 Classwork — Equations & Inequalities

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1

Solve: $x + 9 = 15$

answer

Problem 2

Solve: $4x = 28$

answer

Problem 3

Solve: $3x + 5 = 20$

answer

Problem 4

Solve: $\frac{x}{3} - 2 = 4$

answer

Problem 5

Solve: $-2x + 7 = 1$

answer

B. Distributive Property & Variables on Both Sides

Problem 1

Solve: $2(x + 3) = 16$

answer

Problem 2

Solve: $3(2x - 1) = 15$

answer

Problem 3

Solve: $5x + 4 = 2x + 13$

answer

Problem 4

Solve: $7x - 2 = 3x + 14$

answer

C. Multi-Step Equations

Problem 1

Solve: $3(x + 2) = 2x + 10$

answer

Problem 2

Solve: $4x - 3 = 2(x + 5)$

answer

Problem 3

Solve: $5 - 2x = 3x + 15$

answer

D. Special Cases — No Solution or Infinite Solutions

Problem 1

Solve: $4x + 3 = 4x + 3$ — how many solutions?

answer

Problem 2

Solve: $2x + 5 = 2x - 1$ — how many solutions?

answer

Problem 3

Solve: $3(x + 4) = 3x + 12$ — how many solutions?

answer

E. Linear Inequalities

Problem 1

Solve: $x + 4 > 9$

answer

Problem 2

Solve: $3x \leq 21$

answer

Problem 3

Solve: $-2x > 8$ (careful!)

answer

Problem 4

Solve: $5x - 3 \geq 12$

answer

Problem 5

Solve: $-3x + 1 \leq 10$ (careful!)

answer

F. Word Problems

Problem 1

A taxi charges $3 plus $2 per mile. If the total fare was $17, how many miles were driven?

answer

Problem 2

A gym membership costs $25 to join plus $15 per month. You can spend at most $100. What's the max number of months?

answer